Vivian Wan Wai Fung

BEng Computing — Imperial College London — London / Hong Kong
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EDUCATION

Imperial College London **Sep 2025 – Jun 2028**
BEng Computing *First-Class Honours (Year 1)*

- **Relevant Modules:** Operating Systems, Computer Architecture, Graphs and Algorithms, Computer Systems, Object-Oriented Programming, Databases
- **Societies:** Algorithmic Trading Society, Robotics Society, AI Society, Google Developer Group Academy

Heep Yunn School **Sep 2019 – Jun 2025**

- Ranked 1st in cohort in Mathematics (Calculus and Algebra) and Information and Communication Technology

TECHNICAL SKILLS

- **Programming Languages:** C, Java, Python, Kotlin, Haskell, SQL, ARMv8 Assembly, x86-64 Assembly
- **Technical Knowledge:** Data Structures and Algorithms, Operating Systems, Concurrency, Computer Architecture, Object-Oriented Programming
- **Tools and Technologies:** Git, CI/CD, SQLite, PostgreSQL, Raspberry Pi GPIO, Latex

TECHNICAL PROJECTS

PintOS – Operating Systems Development *C, Git — Ongoing*

- Developing kernel-level functionality in PintOS, focusing on thread scheduling, synchronisation and concurrency.
- Working on priority-aware scheduling and thread synchronisation using locks and semaphores.
- Investigating race conditions and busy waiting through operating-system testing and debugging.

ARMv8 Assembler and Emulator *C, ARM Assembly, Git*

- Co-developed an ARMv8 assembler and instruction-set emulator in C within a four-week team project.
- Implemented reusable data structures and instruction-processing utilities
- Integrated Raspberry Pi GPIO interfaces to control external circuitry through decoded instructions.
- Established automated unit tests and Git-based CI/CD workflows to validate software changes.

Tournament Tree and Chart Generator *Python, SQLite, Tkinter*

- Developed a desktop application to generate tournament brackets and manage chess competition data.
- Designed SQLite database structures and implemented custom pairing and grouping algorithms.
- Built a Tkinter interface with authentication and data validation mechanisms.

LEADERSHIP & ACHIEVEMENTS

Organiser, IC Hack 2026

- Co-organised a major student-run hackathon attended by 700+ participants, coordinating event logistics and sponsor communications.
- Coordinated 100+ volunteers and collaborated with technical teams on website UI design and event delivery.

Finalist, Bending Spoons First Commit East Asia

- Selected as one of 30 engineering student finalists across East Asia for a five-day engineering challenge in Milan (Nov 2026).